Votes and Capital

Obstructions to conversion in a cost-accounted polity, and the yield of the political cycle as the rule of law

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Abstract

Two token colours are put in one cost-accounted rho world: a franchise token V, issued one per registered citizen in a pre-election phase and spendable only at a ballot, and a capital token C, unequally held and spendable everywhere else. Both are linear. The question is when a conversion path from C to V collapses the distinction between what the polity may decide and what money may buy. The answer offered here is that the pathology is not the conversion edge but the *cycle*: capital buys votes, votes buy an office, the office pays rent, and the rule of law is the statement that this closed path has yield at most one. That places the question inside the no-arbitrage apparatus of [4] and the cycle-space error-correction apparatus of [5], and both do real work. Four consequences are drawn out. First, arbitrage removal is *not* the repair: its fixed point is the polity in which a vote is correctly priced at the office's rent per member of the quorum, so the obstruction wanted here is the deletion of an edge, not the adjustment of a rate. Second, the capture cost is a minimum-weight winning coalition, so it is an order statistic of the lower tail of civic virtue and not a function of its mean; with the mean held fixed at 100, dispersion alone moves the capture cost from 5100 to 2076.58 and flips the polity from safe to captured. Third, the first Betti number of the conversion graph counts, with one number, both the audit budget of the price code and the number of independently purchasable offices. Fourth, typing the franchise constitutively and demanding co-authentication at the ballot does obstruct the *sale* of a vote — and converts it into a *delegation* whose cycle yield is larger by exactly the tenure T. Computations in `polity.py`; a three-citizen world in `polity.rho`.

1 Introduction

The preceding notes in this series asked how to *remove* arbitrage [4, 5]. This one asks the opposite question. There are conversions one wants to be impossible. A polity that distributes a franchise equally and then permits it to be exchanged for capital has, in the usual phrase, sold the rule of law; and the interesting formal question is what exactly has to be absent from the world for that sale not to be available, and what the price of its absence is.

The setting is cost-accounted rho [1] with located token stacks [3, 2]: a stack is a message on a channel, and the right to spend from it is exactly the right to receive on that channel. Two colours are in play. The franchise V is issued one per registered citizen in a pre-election phase — a proxy for checking the register — and is accepted only by the ballot. Capital C is held in whatever distribution capital is held in and is accepted by the market. Both are linear: a token is consumed by the receive that reads it, and no rule of the calculus copies one.

The thesis of the note is a single displayed quantity. Let q be the quorum, θ the price of a vote, R the rent the office dispenses per epoch and T its tenure. Then

$$Y \;=\; \frac{RT}{q\theta}$$

is the yield of the closed path $\mathsf{C} \to \mathsf{V}^{*q} \to \mathsf{O} \to \mathsf{C}$, and the rule of law is the statement $Y \leq 1$. Everything else in the note is either a consequence of that statement or an account of what the statement leaves out.

What is imported

From [4]: conversion systems over a signature monoid, the log-rate cochain, arbitrage as non-unit cycle yield, and the equivalence between arbitrage-freeness and the existence of a valuation. That paper works with *reversible* engines; §3 departs from it deliberately, and §9 records the departure.

From [5]: engines as edges, the detection theorem $\|\text{syndrome}\|^2 = a^2\big(1 - R_{\text{eff}}(e)\big)$, Foster's identity $\sum_e (1 - R_{\text{eff}}(e)) = \beta_1$ as the total detection budget of the price code, "bridges fail twice", and the reading of interposition as the general error model of a medium.

From [8]: quorum structures as the opens of a semitopology, and the observation that a coalition is as strong as its weakest member. Here the inequality runs the other way: a polity is as corruptible as its cheapest quorum.

From [9]: the constitutive/possessive distinction, and the mixed-zone (DILL-style) calculus in which reputational hypotheses are unrestricted and value hypotheses linear. §8 is where that distinction earns its keep.

From [3]: located stacks, and the slogan that authority is possession of a name and never adjacency. Everything in §8 turns on it.

What is assumed away

The ballot funds its own metering. In cost-accounted rho a rendezvous burns phlogiston, so a ballot whose gate is paid by the voter has a poll tax built into it and the effective franchise of citizen i is $\min\big(\mathsf{V}_i, \lfloor \sigma_i/\text{gate} \rfloor\big)$. That is a real leak and a different note; here every ballot is free at the point of use.

2 The polity

Definition 2.1 (The world)**.** Fix n citizens $A_1, \ldots, A_n$ and an epoch counter. The world is the parallel composition of

(i) a *roll*, a persistent process answering whether a signature is registered;

(ii) an *issue* phase, which at the start of each epoch places one V token on the located stack $v_i = @[*\text{roll}, e, \Sigma(A_i)]$ for each registered A_i;

(iii) a *ballot*, which consumes V tokens and, on reaching quorum q for one choice, emits the *office* capability;

(iv) a *rent* engine, which pays R units of C per epoch to the holder of the office for T epochs;

(v) a *market*, which consumes C and delivers goods and services.

Appendix A gives all five for $n = 3$, $q = 2$.

Three points about the issue phase deserve emphasis, because the rest of the note leans on them.

Observation 2.2 (The franchise is a source, not a mint)**.** Nothing in the issue phase creates a token that was not already entailed by the roll. In the access-profile classification of [7, §10] the roll is a *source*: persistent, rate-limited, non-exclusive, non-adaptive — and here additionally *keyed to identity*. Capital is *stock*: exhaustible, exclusive, accumulable. The whole of §7 is a consequence of this difference in profile rather than of any difference in quantity.

Observation 2.3 (Two zones, one token)**.** The roll is non-rivalrous: consulting it does not deplete it, and contraction is sound on the hypothesis $\Gamma \vdash \Sigma(A_i) : \mathsf{Citizen}$. The issued token is rivalrous: casting it consumes it. So the franchise is *constitutive in origin and possessive in circulation*, and the issue phase is precisely the dereliction

$$\Gamma \vdash\, !\,\mathsf{Citizen}(A_i) \quad \longmapsto \quad \Gamma\,;\,\Delta \vdash v_i : \mathsf{V}(A_i)$$

metered to one dereliction per epoch. This is the mixed-zone calculus of [9, §6.3] with a rate limit attached to the dereliction rule.

Observation 2.4 (Locations are names)**.** Because $|$ is associative–commutative, a stack that floats in the soup is adjacent to every process in it, which is ambient authority [3, §2.2]. Every stack here is held on a channel. The consequence that matters is negative: the object of a bribe is not the token but the *channel*, and the calculus has no rule that prevents a process from sending a name it holds.

3 The political cycle and its yield

3.1 The conversion graph

Following [4, §2] we work over the signature monoid $(\Sigma, *, ())$ and take the token set $\mathsf{T} = \{\mathsf{C}, \mathsf{V}, \mathsf{O}_1, \ldots, \mathsf{O}_m\}$, closed under $*$ on the fragment it names, so that V^{*q} is the compound signature of a quorum's worth of franchise.

Definition 3.1 (The political engines)**.** Γ_{pol} has three families of engine.

$\varepsilon_i : \mathsf{V}^{*q} \to \mathsf{O}_i,$	$r(\varepsilon_i) = 1$	the ballot: q votes make one tenure;
$\rho_i : \mathsf{O}_i \to \mathsf{C},$	$r(\rho_i) = R_i T_i$	the rent: a tenure pays $R_i T_i$;
$\beta : \mathsf{C} \to \mathsf{V},$	$r(\beta) = 1/\theta$	the bribe: θ capital buys one vote.

Remark 3.2 (Irreversibility, and why the vtok theorem must be restated). Definition 2.1 of [4] imposes $r(a \to b)\, r(b \to a) = 1$, which makes every engine an undirected edge. That is wrong here. An office cannot be converted back into q votes; a delivered service cannot be converted back into capital. The political engines are one-way, and Γ_{pol} is a *directed* graph. The correct restatement is the standard negative-cycle/potential duality, which specialises to [4, Thm. 3.3] when every edge is present in both orientations.

Theorem 3.3 (Sub-valuations and directed cycles)**.** *Let Γ be a finite directed conversion graph with rates $r : E \to \mathbb{R}_{>0}$ and log-rates $\ell = \log r$. The following are equivalent.*

(i) *No directed cycle has yield > 1.*

(ii) *There is a* sub-valuation*: a function $\nu : \mathsf{T} \to \mathbb{R}_{>0}$ with $\nu(a) \geq r(a \to b)\,\nu(b)$ for every engine $a \to b$; equivalently a potential $p = \log \nu$ with $\ell(a \to b) \leq p(a) - p(b)$.*

(iii) *The longest-path problem for ℓ on Γ is well posed.*

Moreover ν may be taken to be a valuation in the sense of [4, Def. 3.2] — with equality throughout — if and only if every directed cycle has yield exactly 1*.*

Proof. (i)⇒(ii): adjoin a source vertex with zero-weight edges to every token and let $p(a)$ be the longest ℓ-weight of a path from the source to a, finite because no positive-weight cycle exists. Relaxation of the edge $a \to b$ gives $p(b) \geq p(a) + \ell(a \to b)$, whence $\ell(a \to b) \leq p(a) - p(b)$ after the sign convention of [4, Def. 3.2]. (ii)⇒(i): telescope p around a cycle; the inequalities sum to $\sum \ell \leq 0$, i.e. yield ≤ 1. (i)⇔(iii) is Bellman–Ford. The last clause is [4, Thm. 3.3] with path-independence replacing path-domination. □

Remark 3.4. The reading of Theorem 3.3(ii) is exactly the reading one wants of a polity: *every available conversion is value non-increasing*. A society in which no sequence of legitimate and illegitimate moves returns more than it consumed is a society with a coherent price for everything, including a price for things that are not for sale. That is a strong statement, and Corollary 3.8 says how strong.

3.2 The theorem

Definition 3.5 (Rule of law)**.** The polity satisfies the *rule of law* when every directed cycle of Γ_{pol} passing through a ballot engine has yield at most one. It is *captured* when some such cycle has yield greater than one and some holder's purse covers its cost.

Theorem 3.6 (The political cycle)**.** *In the minimal polity ($m = 1$, one bribe engine, homogeneous vote price θ) the unique directed cycle through the ballot is $\gamma = \beta^{*q}; \varepsilon; \rho$, with yield*

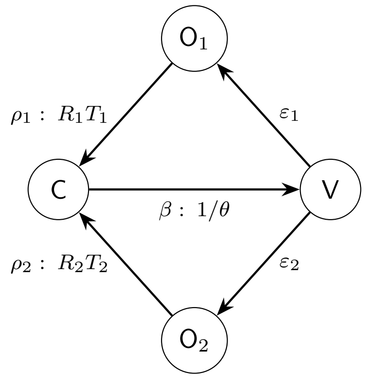
$$Y(\gamma) \;=\; \frac{RT}{q\theta}.$$

Hence the rule of law holds if and only if $q\theta \geq RT$, and the polity is captured if and only if additionally some purse satisfies $W \geq q\theta$.

Proof. Monoidal compatibility [4, Def. 2.2] gives $r(\mathsf{C}^{*q} \to \mathsf{V}^{*q}) = \theta^{-q}$. Composing along γ: $q\theta$ units of C yield q units of V, yield one tenure of O, yield RT units of C. Uniqueness of the cycle is $\beta_1(\Gamma_{\text{pol}}) = 1$ at $m = 1$ (Proposition 6.1). □

Two separate conditions appear, and conflating them is the commonest error in informal discussion of this subject.

Corollary 3.7 (Liquidity and profitability are different)**.** *Capture requires both $RT > q\theta$ (the cycle is profitable) and $W \geq q\theta$ (somebody can front the cost). A polity of many modest fortunes can be safe against a cycle of enormous yield, and a polity with one great fortune can be safe against a cycle of yield just under one. Only the first is a property of the constitution; the second is a property of the wealth distribution, and it moves.*



$\beta_1 = m$, one cycle per office.
Delete $\rho_1, \ldots, \rho_m$ and β
becomes a bridge: no cycle,
no syndrome, no price.

Figure 1: The political conversion graph at $m = 2$. Every cycle passes through β; the first Betti number counts the offices.

Corollary 3.8 (The price of the franchise)**.** *The polity sits exactly on the boundary of Definition 3.5 iff $\theta = RT/q$, and in that case, and only in that case, a valuation exists with*

$$\frac{\nu(\mathsf{V})}{\nu(\mathsf{C})} \;=\; \frac{RT}{q}.$$

Arbitrage-freeness of a polity is therefore not the absence of a price for a vote. It is the assertion that a vote has *one, namely the office's lifetime rent divided by the quorum.*

At $R = 1200$, $T = 4$, $q = 51$ the break-even vote price is $\theta^* = 94.1176$.

Corollary 3.9 (Arbitrage removal is not the repair)**.** *The arbitrage-removal reflection of [4, §4] adjusts rates until the cyclic component of the log-rate cochain vanishes. Applied to Γ_{pol} it raises θ to RT/q: its fixed point is the polity in which vote-buying is break-even rather than the polity in which it is absent. The obstruction sought in this note is therefore* not in the image of the reflection*. It is the deletion of the edge β — a change to the topology of Γ_{pol}, not to its rates.*

Principle 3.10 (Two obstructions, one of which is a price)**.** A polity may obstruct the conversion in exactly two ways. *Topologically*, by ensuring that no capability reaches the franchise namespace from a C-holding one, which deletes β and disconnects V from C; [5, §10] then says there is no common valuation and the two colours are incommensurable. *Economically*, by ensuring $RT \le q\theta$. The second inequality is symmetric in RT and θ, which is the note's most useful normative remark: *bounding what an office may dispense and raising the price of civic virtue are the same intervention*, and only one of the two is under the constitution's control.

4 The capture cost is a minimum-weight winning coalition

Theorem 3.6 assumed one price θ for every citizen. Dropping that assumption is where the model starts to say something.

Definition 4.1 (Quorum structure and capture cost)**.** A *quorum structure* is an upward-closed family $\mathcal{W} \subseteq 2^{\{A_1,\ldots,A_n\}}$ of winning coalitions — in the language of [8], a choice of opens. Given reservation prices $\theta_1, \ldots, \theta_n$, the *capture cost* is

$$K(\mathcal{W}, \theta) \;=\; \min_{S \in \mathcal{W}} \sum_{i \in S} \theta_i.$$

σ	K	$Y = RT/K$	
0.0	5100.00	0.941176	rule of law holds
0.1	4674.95	1.026749	captured
0.2	4255.19	1.128034	captured
0.4	3449.94	1.391328	captured
0.8	2073.11	2.315363	captured
1.2	1098.62	4.369130	captured
1.6	510.07	9.410449	captured

Table 1: Common random numbers; the mean reservation price is held at 100.00 throughout. The crossing $Y = 1$ occurs at $\sigma \approx 0.0687$. The sweep uses its own standard-normal draw, held fixed across rows so that the curve is monotone, so the $\sigma = 0.8$ row differs slightly from the independently drawn $K = 2076.58$ of §4.

Proposition 4.2 (Order statistics)**.** *For the anonymous rule $\mathcal{W} = \{S : |S| \geq q\}$, K is the sum of the q smallest reservation prices. It is therefore a functional of the* lower tail *of the distribution of civic virtue and not of its mean.*

The consequence is sharp enough to be worth measuring. With $n = 101$, $q = 51$, and reservation prices lognormal with the mean pinned at 100:

distribution	K	$Y = RT/K$	verdict
every citizen prices at 100	5100.00	0.941176	rule of law holds
lognormal, mean 100, $\sigma = 0.8$	2076.58	2.311494	captured

Corollary 4.3 (The incorruptible are irrelevant)**.** *In the measured draw the 50 dearest citizens hold 76.1% of the aggregate reservation price of the electorate and contribute nothing whatever to K. Virtue concentrated above the q-th order statistic buys the polity no protection at all.*

This is the mirror image of the quorum algebra of [8, §6], where a coalition is as strong as its weakest member because $\otimes$ is $\min$. Here the polity is as corruptible as its cheapest quorum, and for the same reason: the relevant operation over coalitions is a minimum, and a minimum is blind to everything but its argmin.

Proposition 4.4 (Dispersion alone captures the polity)**.** *Holding the mean reservation price fixed, the capture cost is strictly decreasing in dispersion, and in the measured sweep (Table 1) the yield crosses one at $\sigma \approx 0.0687$ — a coefficient of variation under 7%. The polity is not captured because virtue is cheap on average. It is captured because virtue is* unevenly priced.

4.1 Districting

Proposition 4.5 (Nesting the rule lowers the cost twice)**.** *Replace the anonymous rule by a two-tier rule: 11 districts of sizes $10, 10, 9, \ldots, 9$ summing to 101, a district won by an internal majority, the office won by 6 districts. Then*

(a) the minimal winning coalition shrinks from 51 to 30 citizens, which lowers K even at constant prices; and

purse W	purchasable share
0	50.00%
250	63.60%
500	73.50%
1000	89.45%
1500	99.75%
2000	100.00%

Table 2: 2000 outcomes, alignment drawn per outcome; heterogeneous reservation prices as in §4. Half the outcomes command a natural majority and are free.

(b) *the choice of* which *citizens share a district is a free parameter of the constitution, and choosing it adversarially lowers K again.*

Measured: $K = 2076.58$ under the popular majority; 1205.40 under a random assignment of citizens to districts; 936.30 under a packed assignment that puts a bare majority of the cheapest citizens into each of six districts and pads them with the dearest. The packed figure is 45.09% of the popular-majority price and requires the consent of 30 of 101 citizens, 29.70% of the electorate.

In the semitopological language of [8] this is the observation that $\mathcal{W}$ is a *choice of opens* and that the choice is priced. Gerrymandering is not an abuse of a quorum structure; it is a move within the space of quorum structures, and the model says exactly what the move is worth.

Remark 4.6 (Raising the quorum is not free either). K is increasing in q, so unanimity maximises the capture cost among anonymous rules — and destroys liveness, since one holdout blocks every outcome. This is the safety/liveness trade of [8, §11] in its constitutional dress, and it is why the interesting designs sit strictly between $q = \lceil (n+1)/2 \rceil$ and $q = n$.

5 The purchasable set

The stated target was the condition under which *all* collectively enabled outcomes become purchasable. Two regimes appear, and the difference between them is instructive.

Proposition 5.1 (Anonymity forbids partial capture)**.** *If all reservation prices are equal and the quorum structure is anonymous, then every collectively enabled outcome has the same capture cost K. The purchasable set is therefore empty for $W < K$ and the whole of the collectively enabled set for $W \geq K$: there is no purse that buys some outcomes and not others.*

Corollary 5.2. *Equality of the franchise, which is what makes the polity fair, is also what makes its corruption a phase transition rather than a gradient. A model with graded capture must get the grading from somewhere other than the distribution of votes.*

It gets it from prior alignment. Give each outcome φ a set of natural supporters $S(\varphi)$; then the briber need only buy $q - |S(\varphi)|$ opponents, and cheapest-first:

$$K(\varphi) \;=\; \sum \big\{ \text{the } q - |S(\varphi)| \text{ smallest } \theta_i \;:\; i \notin S(\varphi) \big\},$$

with $K(\varphi) = 0$ when $|S(\varphi)| \geq q$.

Proposition 5.3 (The price of everything)**.** *In the measured world a purse of* $W^* = 1663.38$ *purchases* every *collectively enabled outcome. That is* 80.10% *of the price of a bare quorum and* 34.65% *of a single tenure's rent.*

The last figure is the one to hold on to. The purse that buys the entire legislative output of the polity is a third of what the office pays out in one tenure. The reason is Corollary 4.3 acting twice: an outcome with any natural support needs fewer than q purchases, and the purchases it needs are the cheapest available.

6 Detection

Proposition 6.1 (The Betti number counts the offices)**.** *For* Γ_{pol} *with* m *rent-bearing offices sharing one bribe edge,* $|V| = m + 2$*,* $|E| = 2m + 1$ *and* $\beta_1 = |E| - |V| + 1 = m$*.*

Theorem 6.2 (Foster, imported [5, Thm. 5.4])**.** $\sum_e \bigl(1 - R_{\text{eff}}(e)\bigr) = \beta_1$*: the total detection strength of the price code is the first Betti number, distributed among the edges by geometry.*

Verified numerically for $m = 1, \ldots, 6$ in `polity.py`: the sum equals m to six decimal places in every case.

Proposition 6.3 (The strength of the bribe edge)**.** *In* Γ_{pol} *with unit edge weights,* $1 - R_{\text{eff}}(\beta) = m/(m + 2)$*: it is* $1/3$ *at one office,* $1/2$ *at two,* $3/5$ *at three and* $3/4$ *at six.*

Proof. The m office paths from V to C each have resistance 2 and are in parallel, giving $2/m$; that is in parallel with the unit edge β, giving $R_{\text{eff}}(\beta) = 2/(m + 2)$. □

Corollary 6.4 (The lone drain is invisible)**.** *If the office bears no rent — delete every* ρ_i *— then* β *lies on no cycle. By [5, Thm. 5.3] its syndrome is identically zero, and by [5, Cor. 5.6] its rate is unconstrained by no-arbitrage: any price for a vote extends to a consistent sub-valuation. In a polity with nothing to sell, the price of a vote is both undetectable and undetermined; a citizen may sell the franchise for a drink and no audit will register it, because there is nothing for the audit to be inconsistent* with*.*

Principle 6.5 (Separation of powers is a first Betti number)**.** The audit budget of the price code over Γ_{pol} is exactly the number of independently rent-bearing offices.

That is the pleasant half. The unpleasant half is that the same number counts something else.

Proposition 6.6 (β_1 is two-edged)**.** $\beta_1 = m$ *counts, simultaneously, the audit budget of Theorem 6.2 and the number of independently purchasable offices. Adding an office adds a cycle, and a cycle is both a check and an opportunity.*

The obvious repair — divide one office of rent RT into m offices of rent RT/m — does less than one hopes.

Proposition 6.7 (Division of the office)**.** *If the* m *divided offices are elected by the* same *electorate, one purchased quorum wins all of them at once, the capture cost is unchanged, and the aggregate yield is unchanged at* RT/K*: division buys audit budget and nothing else. If they are elected by* disjoint *blocks, the cost of total capture rises — measured* $2076.58 \to 2267.96 \to 2402.77 \to 2691.25$ *for* $m = 1, 3, 5, 11$ *— because the briber may no longer cherry-pick globally; but the yield of the* cheapest single office *does not fall, measured* $2.311494, 2.470722, 2.361207, 2.332065$*. Partial capture remains exactly as profitable as total capture was.*

Remark 6.8 (A labelled negative result). Proposition 6.7 is stated as a negative result and kept. The tidy story — separate the powers and the yield divides — is false in this model, and the reason is structural rather than numerical: the yield of a cycle is a ratio, and dividing the rent divides the numerator and the number of purchases equally.

Remark 6.9 (The honest caveat about detection). Theorem 6.2 prices *rate faults on observable edges*. The bribe edge is the least observable edge in the graph: reading it requires grade-one access to a private rendezvous, which in the classification of [7, §2] is exactly what an outsider does not have. So the realised detection strength is β_1 scaled by transparency, and a constitution that multiplies offices while leaving β unobservable has bought a budget it cannot spend. §8 approaches this from the other side and finds one mechanism that makes the corrupt edge visible — at a price.

7 The ratchet

Proposition 7.1 (A stock cannot be outbid by a flow)**.** *The franchise is re-issued each epoch and is not accumulable: an unspent* V *token is void at the next registration. Capital persists. Hence a defensive counter-bid — citizens outbidding a briber for their own votes — must be funded from stock, and the electorate's stock is the thing capital exists to spend, while the incumbent's stock is fed by* ρ*.*

Measured, with $K = 2076.58$, $RT = 4800$, net 2723.42 per tenure, against an electorate whose combined capital stock is $49{,}108.98$: the incumbent's stock passes the whole electorate's at tenure 18, that is after 72 epochs, over which 7272 votes were issued and none of them stored.

Corollary 7.2. *The asymmetry that drives capture is between* access profiles *— source against stock — and not between quantities. Equalising the initial distribution of capital would delay the crossing; it would not change its direction, because only one side of the exchange compounds.*

8 Sale, delegation, and the price of co-authentication

The remaining question is whether typing the franchise correctly obstructs the edge β at all. The answer is a qualified yes with an unwelcome quantitative tail.

Proposition 8.1 (Anonymity of send; constitutive typing alone obstructs nothing)**.** *A rho send carries no sender. A ballot whose guard consumes the token* $\mathsf{V}(A_i)$ *and inspects its content therefore cannot distinguish a cast by* A_i *from a cast by any process holding the located stack channel* v_i*. Consequently the constitutive* typing *of the franchise — the fact that the token records whose franchise it is — obstructs nothing on its own: a briber holding* q *distinct tokens casts* q *distinct identities and the ballot is satisfied.*

Definition 8.2 (Co-authenticating ballot)**.** The *co-authenticating* ballot's guard demands two things: the linear token $\mathsf{V}(A_i)$, and a *fresh attestation* obtained by drawing on the roll at cast time, i.e. a witness for $\Gamma \vdash \Sigma(A_i) : \mathsf{Citizen}$ produced during this ballot.

Theorem 8.3 (Sale collapses to delegation)**.** *Under the co-authenticating ballot, no transfer of linear tokens enables a cast. To cast on behalf of* A_i *a briber must obtain a capability that draws on the roll. Because the roll is unrestricted, that capability is not consumed by use: it is a* standing delegation*, persistent in the term as an interposed forwarder on the path to* v_i*.*

Proof. The guard is satisfiable only by a message produced by the attestation contract, which consults the roll. The roll answers on the signature, and the signature is not a transferable resource in

Δ but a subject position in Γ; so the briber's only route is to hold a name whose receipt triggers the attestation. Holding such a name is possession of a capability drawing on an unrestricted hypothesis, on which contraction is admissible; hence it survives use. $\square$

Corollary 8.4 (Visibility). *A sale leaves nothing behind: the token is consumed and the ballot's transcript is identical to an honest one. A delegation leaves a persistent forwarder, which is an interposed medium in the sense of [5, §5] and therefore a spatial-behavioural formula an auditor can look for. Co-authentication buys the observability that Remark 6.9 said the audit budget needed.*

Theorem 8.5 (The price of visibility). *Because V is re-issued at every registration, a sale must be repurchased every epoch, costing TK over a tenure; a delegation is bought once per tenure, costing K. Hence*

$$Y_{\text{del}} \;=\; T \cdot Y_{\text{sale}}.$$

Measured at $T = 4$: the sale arrangement costs 8306.32 per tenure and yields 0.577873 — the polity is safe. The delegation arrangement costs 2076.58 and yields 2.311494 — the polity is captured. The ratio of yields is 4.000000, exactly the tenure. The delegation cycle becomes profitable as soon as $T > K/R = 1.7305$.

Principle 8.6 (Co-authentication trades expense for visibility). Demanding co-authentication at the ballot converts an invisible, expensive corruption into a visible, cheap one, with the discount factor equal to the tenure. Whether that is a good trade depends entirely on whether anyone is looking; and Remark 6.9 shows that "anyone looking" is itself a budgeted quantity.

Remark 8.7 (This is what one observes). The corrupt relationships that survive in polities with authenticated ballots are exactly the durable, visible, cheap ones: party machines, whipped blocs, custodial staking, delegated proof of stake. The framework predicts that shape from the metering discipline alone — the linear zone is re-issued per epoch, the unrestricted zone is not — and does not need a story about human motives to do it.

9 What is shown and what is not

Shown. That the pathology is a cycle and not an edge (Theorem 3.6); that arbitrage removal prices the franchise instead of protecting it (Corollary 3.9); that the capture cost is an order statistic of the lower tail and that dispersion alone flips the verdict (Propositions 4.2, 4.4); that anonymity makes capture a phase transition and alignment supplies the gradient (Proposition 5.1, Table 2); that β_1 counts audit budget and purchasable offices with one number (Propositions 6.1–6.6); and that co-authentication converts sale into delegation at a yield multiple of T (Theorems 8.3, 8.5).

Not shown.

- *The reservation prices are exogenous.* A citizen's θ_i ought to depend on what the office will do with the purchased outcome, which makes θ a function of K and the whole system a fixed point. Nothing here solves that fixed point.
- *Coercion is not distinguished from sale.* Both appear in the model as a transfer at price θ_i; a coerced transfer has θ_i negative in the citizen's own accounting, which the present bookkeeping cannot represent.

- *The secret ballot is deliberately absent.* With the hidden-name quantifier retired [7, §3], concealment in this framework is a price and never a wall, so a secret ballot would enter as an *expense on the buyer's assay* — the buyer cannot verify delivery, so the purchased good is observable only at a budget-relative $\perp$. That is the natural next note and it is not attempted here.

- *Reversibility was dropped by hand.* Remark 3.2 restates [4, Thm. 3.3] for directed graphs; the restatement is standard, but the Hodge decomposition of [4, §4] does not transport to the directed setting without work, and Corollary 3.9 therefore appeals to the reflection in its reversible form.

- *No dynamics.* Everything is per-epoch bookkeeping. Running the polity under the Gillespie discipline of [10] would let the ratchet of §7 be a rate rather than a recurrence.

10 Open problems

1. **Endogenous virtue.** Solve the fixed point $\theta = \Theta(K(\theta))$. Does it have a stable solution, and is the stable one above or below RT/q?

2. **A coreflection for edge deletion.** Arbitrage removal is an idempotent reflection on rates. Is there a companion operation on *topology* — deletion of the minimal set of edges disconnecting two colours — and is it a coreflection? Principle 3.10 says the two obstructions are different in kind; a categorical statement of the difference would be worth having.

3. **The secret ballot as an engineered $\perp$.** Formalise unverifiability of delivery as a budget-relative bottom in the assay trichotomy of [7, §4], and compute what it does to Y.

4. **Three colours.** Add reputation as a third, non-transferable colour and ask which conversions among $\{\mathsf{V}, \mathsf{C}, \mathsf{Rep}\}$ must be absent. [9] suggests the interesting edges are the ones into and out of the unrestricted zone.

5. **Learning the quorum structure.** [8] makes the semitopology the thing an observer is guessing. Here $\mathcal{W}$ is given by the constitution — but a briber must *discover* which coalitions are cheap, and that search is itself priced. What does the ladder look like?

6. **Does the ratchet have a critical rent?** Below what R does the incumbent's stock fail to compound faster than the electorate's spends down?

A The three-citizen world

`polity.rho` gives the five components of §2 for $n = 3$, $q = 2$, together with the forbidden exchange contract (commented out, since its presence is the thing under study) and the co-authenticating ballot with one delegation. Three details are worth pointing at.

A.1 Located franchise. The issue phase writes to $v_i = @[*\text{roll}, e, \Sigma(A_i)]$. The name is manufactured by quotation and is fresh in the sense of [7, §3] — no `new` binder is needed, and none is used.

A.2 The guard is a ballot, not a market. The guard `where tok == (colour, voter, e) and colour == "V"` is what makes the contract a ballot: it accepts one colour and rejects the other. There is no sender to check, which is Proposition 8.1 in one line of code.

A.3 The forbidden edge is a capability transfer. The commented `exchange` contract does not move a token. It moves the right to receive on v_i. Nothing in rho forbids sending a name one holds, which is why Principle 3.10's topological obstruction is a statement about who was ever handed what, and not a rule of the calculus.

B The computations

`polity.py` is standard library only, seeded at 20260823, and prints every number quoted above to `polity-output.txt`. Sections M1 through M9 correspond to §6, §3, §4 (three sections), §5, §7, §8 and Proposition 6.7 respectively. Effective resistances are computed by grounding one vertex of the graph Laplacian and solving by Gaussian elimination with partial pivoting; Foster's identity is then a check on the implementation as well as a result, and it holds to six decimal places for $m = 1, \ldots, 6$.

Parameters throughout: $n = 101$, $q = 51$, $R = 1200$ per epoch, $T = 4$ epochs, $RT = 4800$, mean reservation price 100, $\sigma = 0.8$ unless a sweep says otherwise.

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